DESCRIPTIVE ANALYTICS REPORT

Zepto

Order-Level Sales, Profitability & Customer Behavior

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Note on Data

This report uses the supplied Zepto sales-order dataset containing 200 order records spanning 1 January 2024 to 18 July 2024 (one order per calendar day). Each order records a single product line: order/customer identifiers, segment, region, category, product, quantity, unit price, discount %, gross/net sales, cost and profit, and payment method.

The dataset shows several structural patterns that are worth flagging before drawing conclusions, since they affect how the KPIs below should be read:

- **Quantity–Region–Payment linkage:** Quantity, Region and Payment Method move in lockstep — every North order has quantity 1 and is paid by Credit Card, every South order has quantity 2 and is paid by PayPal, and so on through East/Debit Card (3), West/Cash on Delivery (4) and Central/Bank Transfer (5). Regional and payment-method comparisons in this report should therefore be read as a reflection of this fixed order-quantity cycle rather than independent regional or payment behaviour.
- **Customer-name cycling:** Only 20 distinct customer names recur across the 200 orders (10 orders each), even though every order carries a unique Customer ID. “Top customer” figures in this report therefore reflect a repeating name cycle rather than confirmed repeat-purchase history.
- **Fixed pricing & balanced category mix:** Each of the 10 products carries a single fixed unit price with no price variation across orders, and each of the 5 categories contains exactly two products and 40 orders — an evenly balanced design rather than an organic sales mix.
- **Two-tier discounting:** Discounting is binary: every order carries either 0% or a flat 10% discount, split exactly 100/100 across the dataset.

Given these patterns, this report treats the dataset as a structured order-level sample suitable for illustrating descriptive-analytics KPIs and reporting mechanics, rather than as organic real-world purchasing behaviour. The analysis below describes what the supplied data shows.

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1. Objective

Build a descriptive analytics layer over the Zepto order-level sales data to answer the following recurring business questions:

- How much revenue, discount and profit did the business generate over the period, and at what margin?
- How is revenue distributed across product categories and individual products?
- How does performance vary by region, customer segment and payment method?
- How did net sales and profit trend month over month?
- What share of orders are discounted, and what is the revenue impact of discounting?
- Which categories and products are the most profitable, by margin as well as by absolute profit?
- How is order value distributed, and what does a typical order look like?
- What does an overall sales-health summary look like?

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2. How This Would Work at Real Scale

Processing a continuous stream of order transactions is an engineering problem before it is an analytics problem. A realistic architecture would ingest order events as they are placed, aggregate them by category, region and time period, retain both raw and processed data, and expose the resulting KPIs through a data-visualization layer.

Layer	Purpose	Typical Tools
Investigation	Research, understand and prepare the order dataset	Google, Claude & ChatGPT
Stream processing	Aggregate by category, region and time period	Jupyter Notebook (Python)
Storage	Retain raw and processed order data	Excel files / CSV files
Batch aggregation	Daily/periodic revenue, discount and margin metrics	Jupyter Notebook
Report	Present analysis, graphs and insights	Data Visualization / MS Word

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3. Sales Overview

Across the supplied dataset, 200 orders were placed between 1 January and 18 July 2024, moving 600 total units for ₹33,300 in gross sales. After ₹1,665 in discounts, net sales stood at ₹31,635, with ₹21,180 in cost of goods and ₹10,455 in profit — an overall margin of approximately 33.0%.

KPI: Overall Sales Summary	
Total orders	200
Total units sold	600
Total gross sales	₹33,300
Total discount given	₹1,665
Total net sales	₹31,635
Total cost	₹21,180
Total profit	₹10,455
Overall profit margin	≈ 33.0%
Average order value (net)	≈ ₹158
Average discount rate	5.0% (blended across 0%/10% orders)

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4. Net Sales & Profit by Category

The chart below shows how net sales are distributed across the five product categories, each represented by two products and 40 orders in the supplied data.

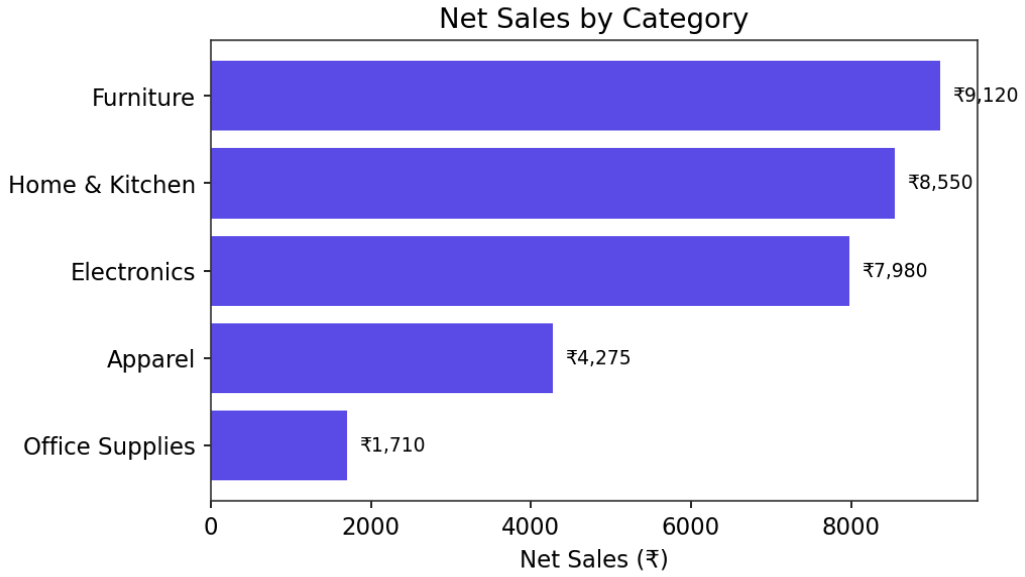


Figure 1. Net sales by category

- Furniture is the top category by net sales (₹9,120), driven by the higher unit price of Monitor Arm and Desk Lamp.
- Home & Kitchen (₹8,550) and Electronics (₹7,980) follow closely behind.
- Office Supplies generates the least revenue (₹1,710) but the highest margin (≈47.4%), reflecting its low cost base relative to price.
- All five categories carry an identical order count (40) and unit volume (120) in this dataset, so revenue differences here are driven entirely by unit price, not by order frequency.

KPI: Category Leaders	
Top category by net sales	Furniture (₹9,120)
Lowest category by net sales	Office Supplies (₹1,710)
Highest-margin category	Office Supplies (≈47.4%)
Lowest-margin category	Furniture (≈30.9%)

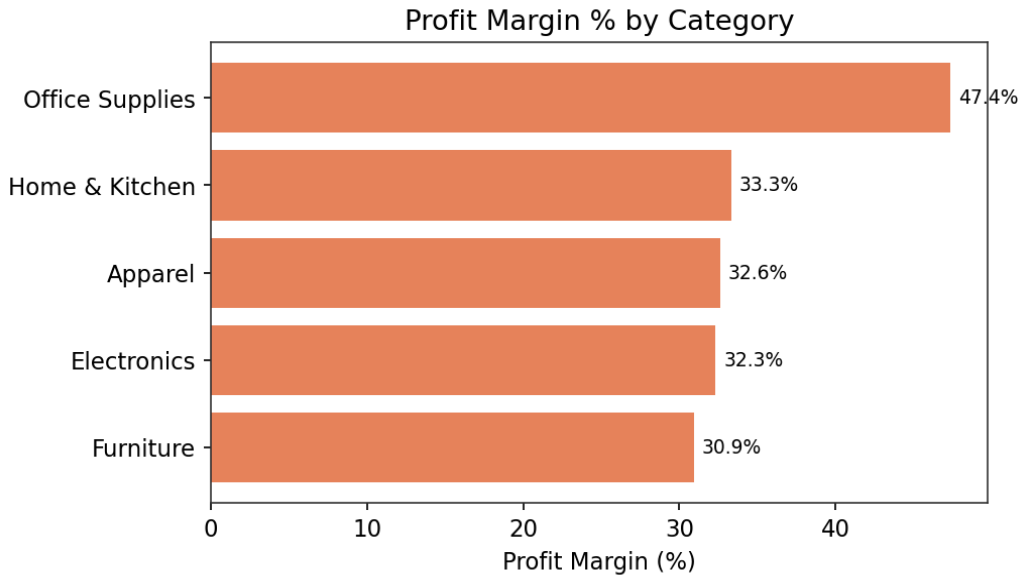


Figure 2. Profit margin % by category

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5. Net Sales by Product

Each of the 10 products in the catalog received exactly 20 orders. The chart below ranks them by net sales.

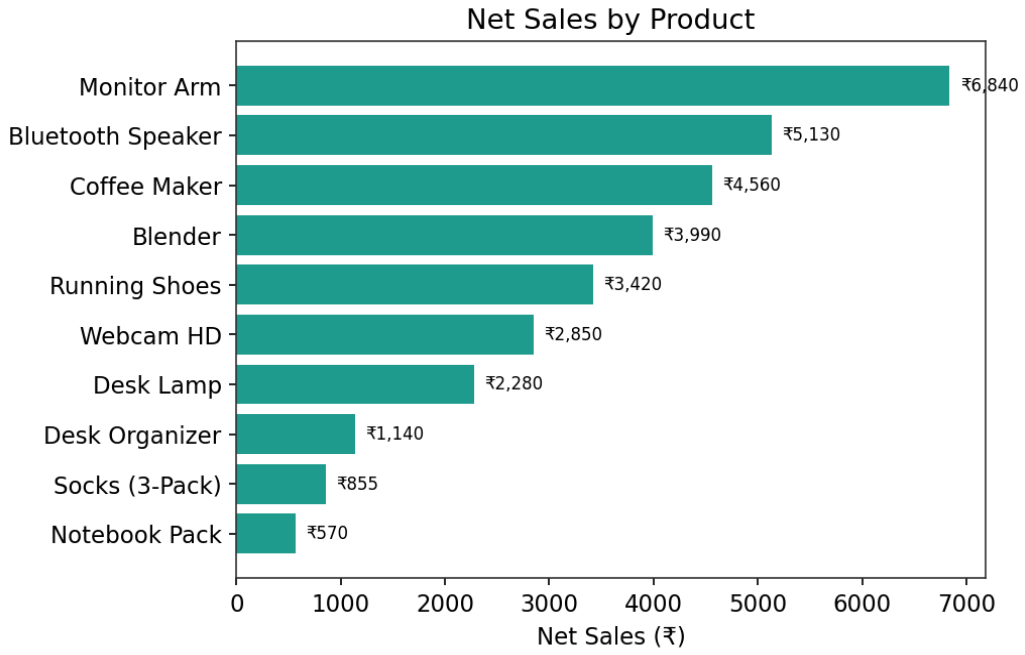


Figure 3. Net sales by product

- Monitor Arm is the single best-selling product by revenue (₹6,840), followed by Bluetooth Speaker (₹5,130) and Coffee Maker (₹4,560).
- Notebook Pack is the lowest-revenue product (₹570), consistent with its ₹10 unit price.

- Webcam HD stands out for margin rather than volume, posting the highest product-level margin among mid-priced items (≈36.8%).

Product	Category	Orders	Units	Net Sales (₹)	Profit (₹)	Margin %
Monitor Arm	Furniture	20	60	6,840	2,040	29.8%
Bluetooth Speaker	Electronics	20	60	5,130	1,530	29.8%
Coffee Maker	Home & Kitchen	20	60	4,560	1,560	34.2%
Blender	Home & Kitchen	20	60	3,990	1,290	32.3%
Running Shoes	Apparel	20	60	3,420	1,020	29.8%
Webcam HD	Electronics	20	60	2,850	1,050	36.8%
Desk Lamp	Furniture	20	60	2,280	780	34.2%
Desk Organizer	Office Supplies	20	60	1,140	540	47.4%
Socks (3-Pack)	Apparel	20	60	855	375	43.9%
Notebook Pack	Office Supplies	20	60	570	270	47.4%

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6. Performance by Region

As noted in the data note above, region is tied one-to-one with order quantity in this dataset (North=1 unit, South=2, East=3, West=4, Central=5 units per order), so the pattern below reflects that fixed relationship rather than independent regional demand.

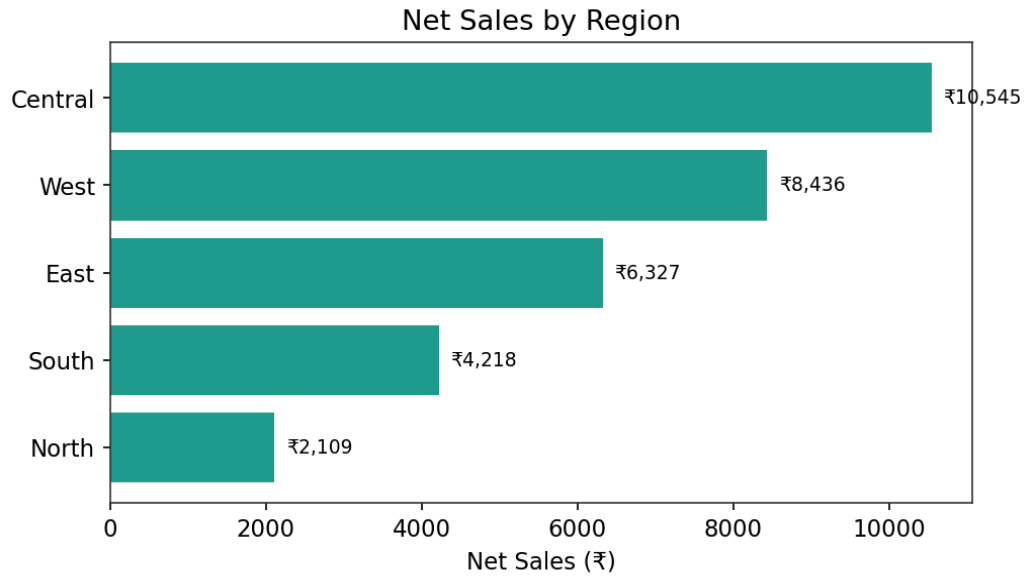


Figure 4. Net sales by region

- Central leads all regions in net sales (₹10,545) and profit (₹3,485), consistent with it always carrying the largest order quantity (5 units) in this dataset.
- North trails all regions (₹2,109 net sales), consistent with its fixed 1-unit order quantity.
- All five regions post an identical profit margin (≈33.0%), since margin is a function of product mix and discount rather than region-specific pricing in this data.

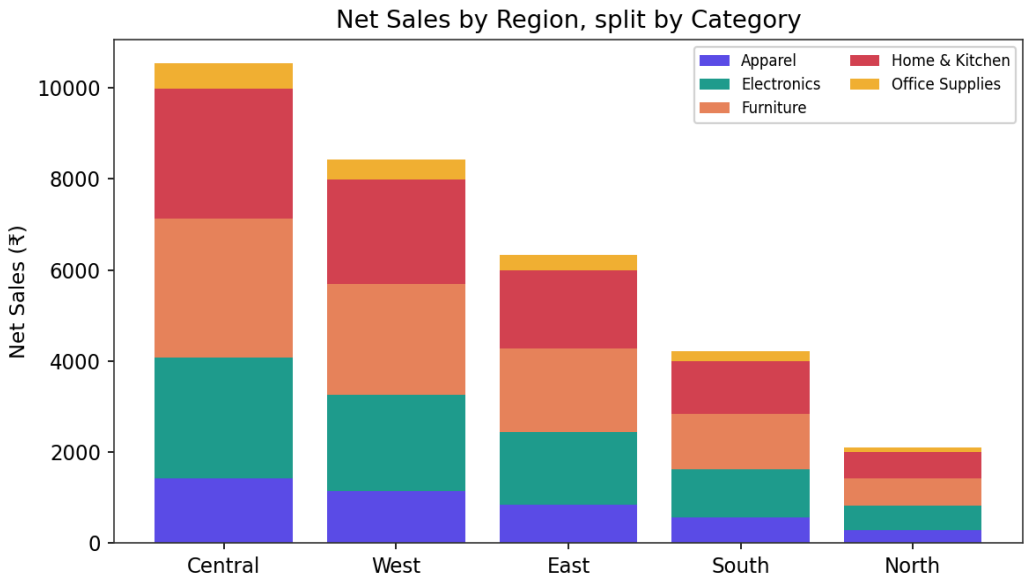


Figure 5. Net sales by region, split by category

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7. Performance by Customer Segment

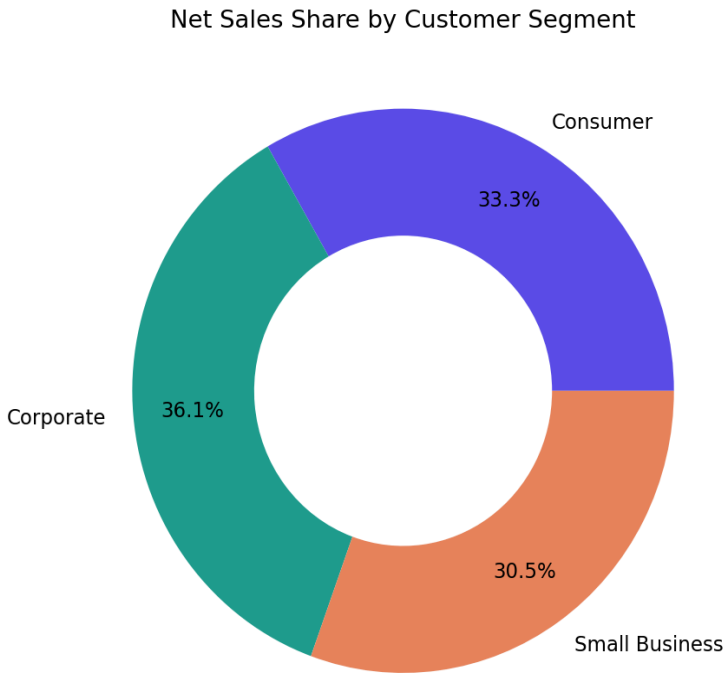


Figure 6. Net sales share by customer segment

- Corporate customers generated the most net sales overall (₹11,433, 70 orders), narrowly ahead of Consumer (₹10,545, 70 orders).
- Small Business placed fewer orders (60) but converted them at the highest margin of the three segments (≈34.2%).

- Corporate carries the lowest segment margin (≈32.1%), suggesting its order mix skews toward lower-margin categories.

KPI: Segment Summary	
Top segment by net sales	Corporate (₹11,433)
Top segment by margin	Small Business (≈34.2%)
Segment order split	Corporate 70 / Consumer 70 / Small Business 60

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8. Performance by Payment Method

Payment method is likewise tied to region and order quantity in this dataset (see the data note), so the split below should be read with that structural link in mind.

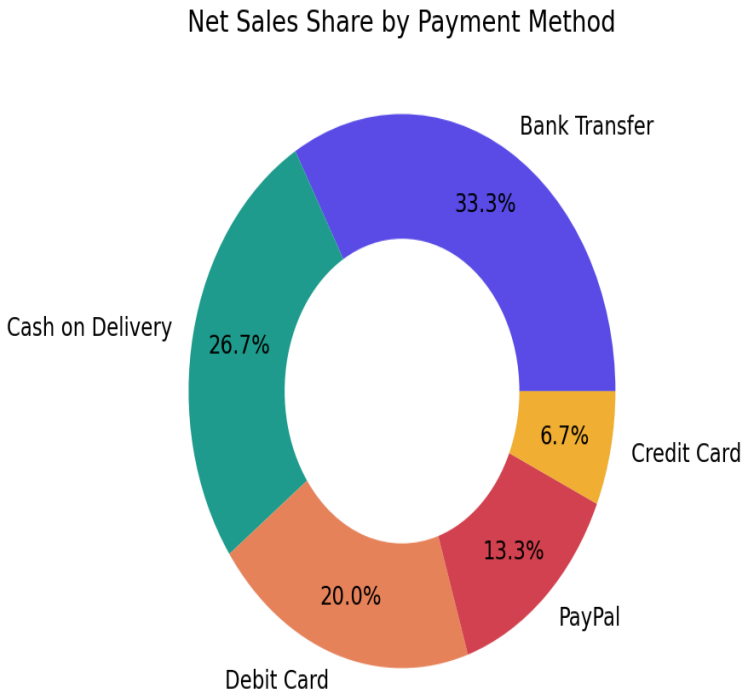


Figure 7. Net sales share by payment method

- Every payment method accounts for exactly 40 of the 200 orders (20% each), but Bank Transfer leads in net-sales value (₹10,545) because it is always paired with the largest order quantity (Central region, 5 units).
- Credit Card carries the smallest net-sales share (₹2,109) for the same structural reason — it is always paired with the smallest order quantity (North region, 1 unit).
- Because order count is identical across methods, this view is more useful here for revenue-per-method context than for judging genuine payment-method preference.

Payment Method	Orders	Share %	Net Sales (₹)	Fixed Region / Qty
Bank Transfer	40	20%	10,545	Central / 5 units

Payment Method	Orders	Share %	Net Sales (₹)	Fixed Region / Qty
Cash on Delivery	40	20%	8,436	West / 4 units
Debit Card	40	20%	6,327	East / 3 units
PayPal	40	20%	4,218	South / 2 units
Credit Card	40	20%	2,109	North / 1 unit

9. Monthly Sales & Profit Trend

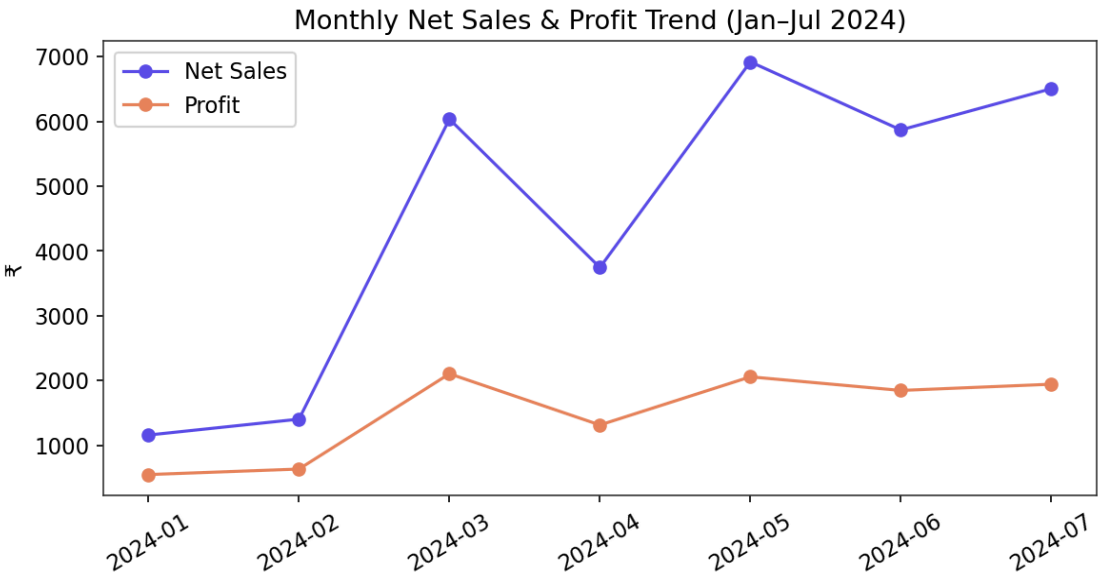


Figure 8. Monthly net sales and profit trend, Jan–Jul 2024

- Net sales rose sharply from January (₹1,160) into March (₹6,035), coinciding with the catalog mix shifting into higher-priced Home & Kitchen and Electronics products.
- May was the strongest month by net sales (₹6,918) and profit (₹2,058) in the recorded window.
- July shows only 18 days of data (through 18 July 2024) yet already recorded ₹6,504 in net sales, suggesting the partial month was tracking close to May/June levels.
- January and February — dominated by the lowest-priced Notebook Pack and Desk Organizer products — are the two weakest months in the dataset.

Month	Orders	Net Sales (₹)	Profit (₹)
Jan 2024	31	1,160	550
Feb 2024	29	1,405	635
Mar 2024	31	6,035	2,105
Apr 2024	30	3,745	1,315
May 2024	31	6,918	2,058
Jun 2024	30	5,868	1,848
Jul 2024 (partial, 18 days)	18	6,504	1,944

10. Discounting & Order-Value Distribution



Figure 9. Orders by discount status

- Exactly half of all orders (100 of 200) carry no discount, while the other half carry a flat 10% discount — there is no intermediate discount tier in this dataset.
- Full-price orders account for ₹16,650 in net sales versus ₹14,985 from discounted orders, a modest revenue gap given the even order split.
- Total discount value given across the period was ₹1,665, about 5.0% of gross sales.

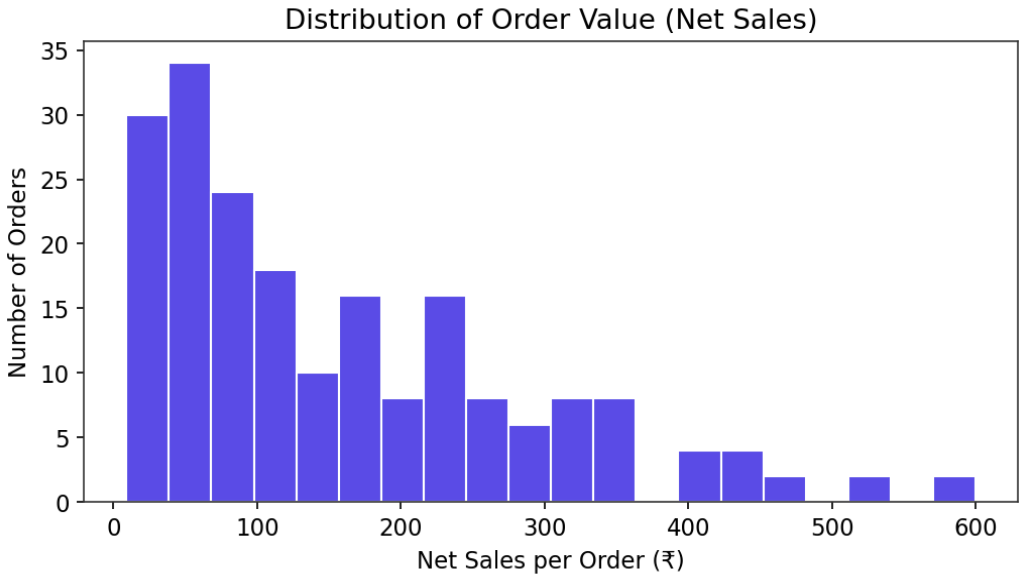


Figure 10. Distribution of net order value

- Order values cluster at the low end (many orders under ₹150), consistent with the mix of low-priced categories like Office Supplies and Apparel.
- A smaller number of high-value orders extend the distribution's tail, driven mainly by Furniture and Electronics orders placed at quantity 4–5.

KPI: Discounting Summary

No-discount orders	100 (50.0%) — ₹16,650 net sales
Discounted orders (10%)	100 (50.0%) — ₹14,985 net sales
Total discount value	₹1,665 (≈5.0% of gross sales)

11. Gross Sales vs Profit Relationship

The scatter plot below relates each order's gross sales to its resulting profit, colored by discount percentage, to show how discounting behaves across order sizes.

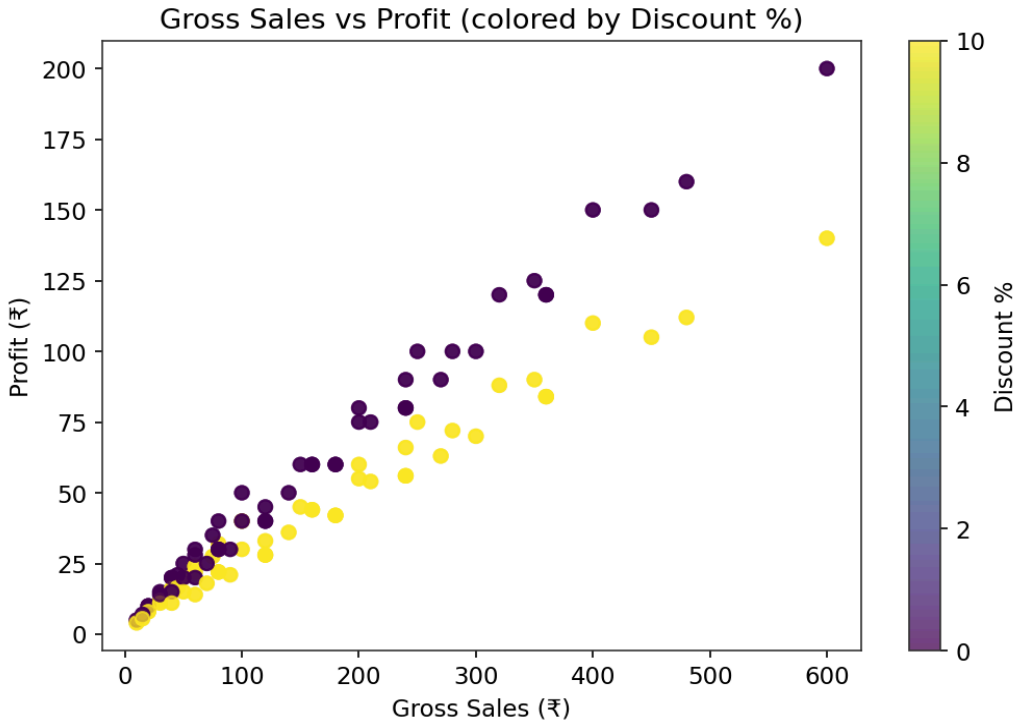


Figure 11. Gross sales vs profit, colored by discount %

- Profit rises broadly in line with gross sales, as expected since profit is derived from the same order value.
- The two distinct color bands (0% and 10%) confirm the binary discount structure — there is no gradient of intermediate discount levels in this data.
- At a given gross-sales level, 10%-discounted orders sit only marginally below full-price orders in profit, since the discount is applied to price rather than cost.

11b. Cumulative Sales Growth

Viewing net sales on a cumulative basis makes the pace of revenue accumulation across the seven-month window easier to read at a glance.

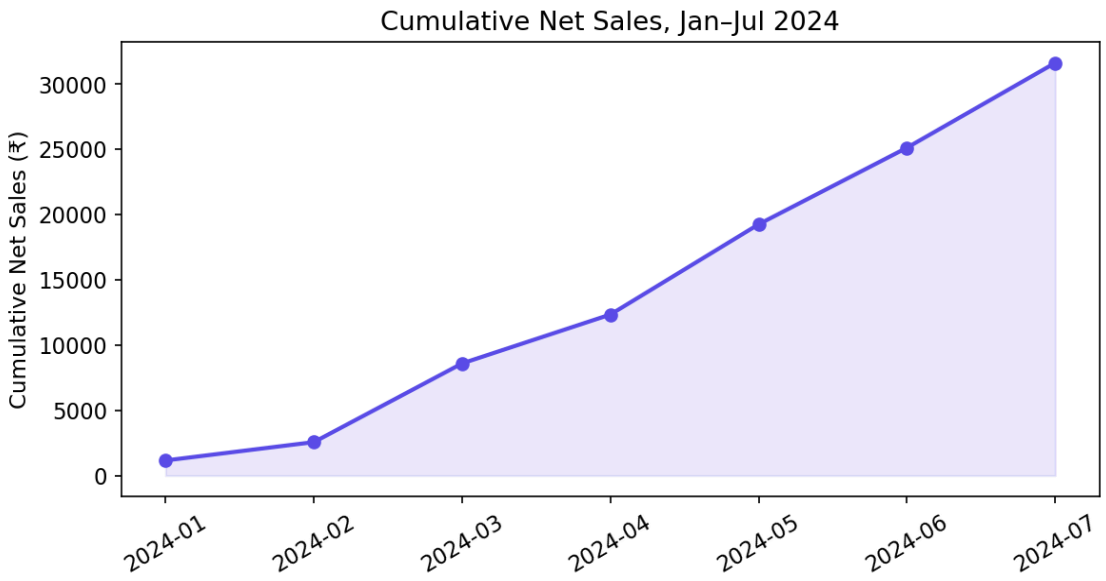


Figure 12. Cumulative net sales, Jan–Jul 2024

- The curve is flat through January–February (≈₹2,565 cumulative), then steepens sharply from March onward as higher-priced categories enter the order mix.
- Roughly 80% of total net sales (₹25,131 of ₹31,635) had already accumulated by the end of June, with the partial July window adding the final ≈20%.
- The steepening slope from March suggests the order mix — not order count, which stays roughly flat at 28–31 per month — is the primary driver of revenue growth in this dataset.

11c. Order Timing: Day-of-Week Pattern

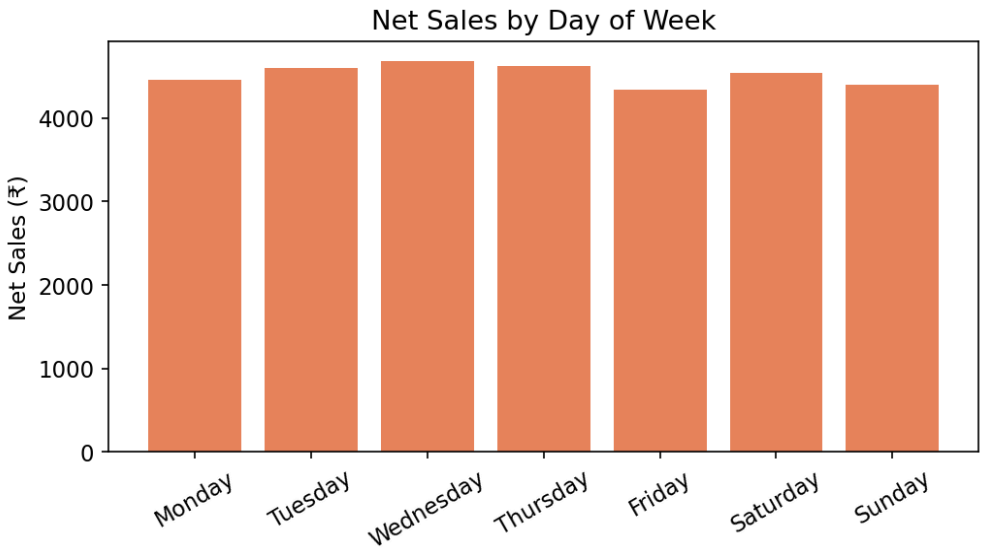


Figure 13. Net sales by day of week

- With one order recorded per calendar day across the 199-day window, each weekday is represented 28–29 times, and net sales are correspondingly even across the week (₹4,336–₹4,683).

- Wednesday is marginally the strongest day (₹4,683) and Friday the softest (₹4,336), but the spread is narrow (≈8%) and not large enough to indicate a meaningful weekday effect in this dataset.
- A genuine day-of-week pattern would need multiple orders per day to separate day-effects from the underlying product/quantity cycle described in the data note.

12x. Order Quantity & Cost Structure

Two further structural views round out the descriptive picture: how order quantity is distributed, and how cost behaves as a share of unit price across products.



Figure 14. Order quantity distribution

- Order quantity is perfectly uniform — exactly 40 orders at each of 1, 2, 3, 4 and 5 units — consistent with the fixed region-quantity cycle noted at the start of this report.
- Because quantity never exceeds 5 units per order, all revenue growth across categories comes from unit price differences rather than bulk ordering.

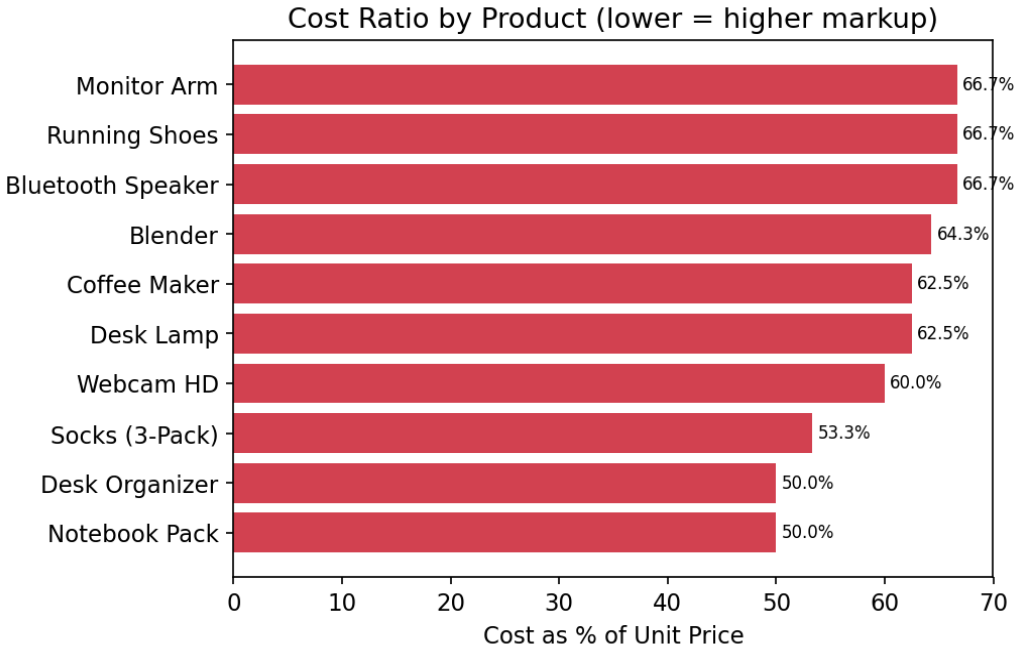


Figure 15. Cost as % of unit price by product

- Notebook Pack and Desk Organizer carry the lowest cost ratio (50% of price), explaining why Office Supplies posts the highest category margin in this report.
- Monitor Arm, Running Shoes and Bluetooth Speaker sit at the other end (≈66.7% cost ratio), leaving the thinnest per-unit markup of the ten products.
- Cost ratio is fixed per product in this dataset (no cost variation across orders of the same item), so margin differences at the category level are entirely a function of product mix.

12y. Category Performance by Customer Segment

Breaking category-level net sales out by customer segment shows which segment contributes most within each category.

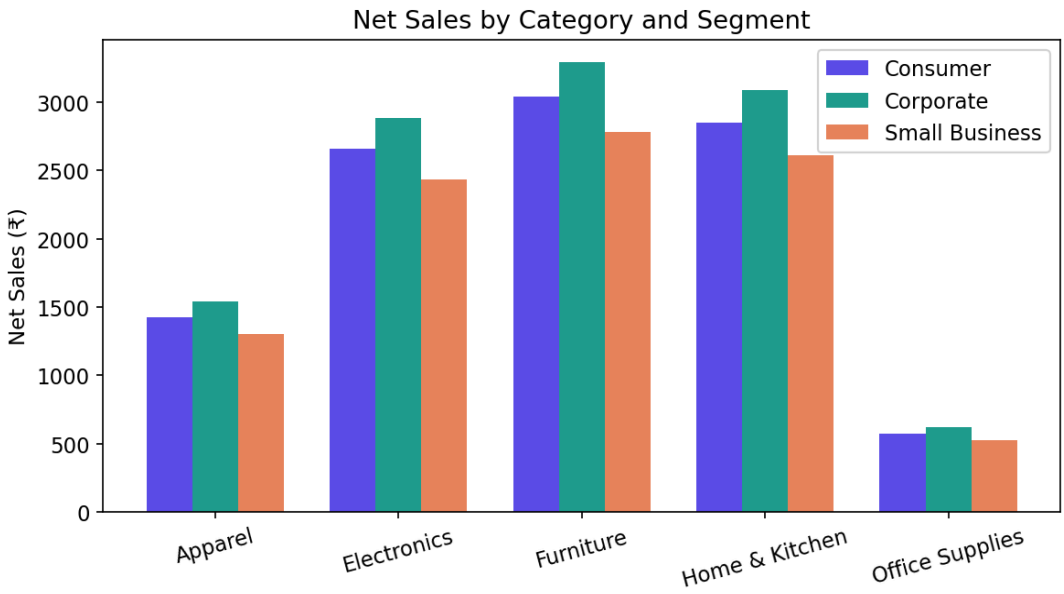


Figure 16. Net sales by category and segment

- Corporate is the leading segment within every one of the five categories, consistent with it holding the most orders (70) among the three segments.
- The gap between segments is proportionally similar across categories (Corporate > Consumer > Small Business in absolute terms), since segment order counts — not category-specific preferences — drive most of the difference here.

Category	Consumer (₹)	Corporate (₹)	Small Business (₹)
Apparel	1,425	1,545	1,305
Electronics	2,660	2,884	2,436
Furniture	3,040	3,296	2,784
Home & Kitchen	2,850	3,090	2,610
Office Supplies	570	618	522

12z. Discount Value & Average Order Value by Category

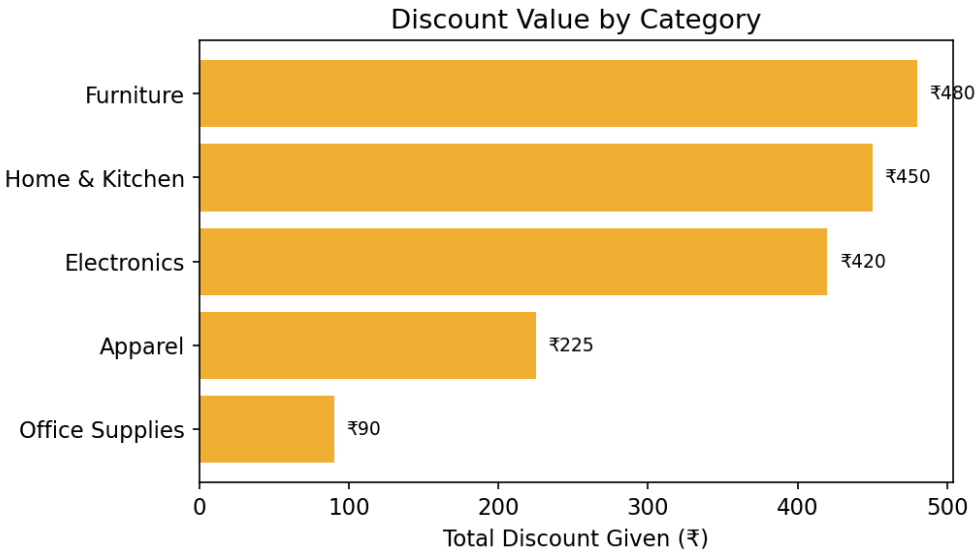


Figure 17. Total discount value by category

- Furniture received the most discount value in absolute terms (₹480), simply because it combines a higher unit price with the same 50% discounted-order rate as every other category.
- Office Supplies received the least discount value (₹90), proportionate to its low unit price.

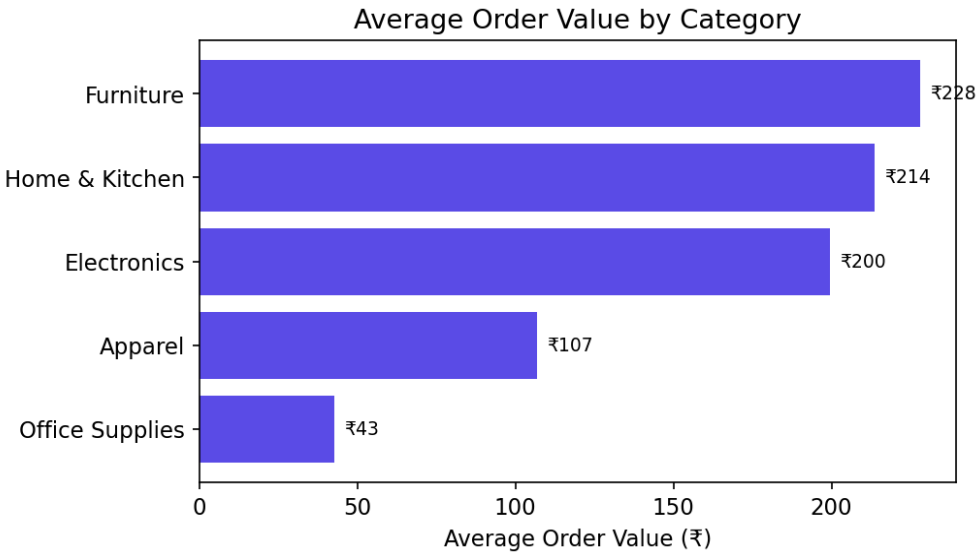


Figure 18. Average order value by category

- Furniture posts the highest average order value (₹228), more than five times Office Supplies' average (₹43).
- Across the full dataset, the median order value is ₹120 and the average is ≈₹158 — the gap indicates a modest right-skew from higher-value Furniture and Electronics orders.

KPI: Order Value Spread	
Minimum order value	₹9
Median order value	₹120
Mean order value	≈₹158

Maximum order value	₹600

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12. Highest-Value Customers

As flagged in the data note, only 20 distinct customer names recur across the 200 orders (10 orders per name), so the ranking below reflects that repeating name cycle rather than confirmed individual repeat-purchase behaviour.

Customer Name	Orders (cycle count)	Total Net Sales (₹)
Michael Brown	10	2,775
Joseph Allen	10	2,775
Nancy Green	10	2,497.5
Mary Rodriguez	10	2,497.5
Christopher Scott	10	2,220
James Martinez	10	2,220
Neha Sharma	10	1,998
Jessica Hall	10	1,998
Arjun Patel	10	1,665
Charles White	10	1,665

- Michael Brown and Joseph Allen top the list (₹2,775 each) — both are always paired with the Central region / 5-unit order quantity in the recurring cycle.
- All top-10 names appear in pairs with identical totals, a direct consequence of the fixed region-cycle structure rather than differing purchase behaviour.

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13. Region Performance by Customer Segment

Cross-tabulating region and segment shows which segment dominates net sales within each region. Because region is itself tied to order quantity (see the data note), the differences below combine both the quantity cycle and each region's segment mix.

Region	Consumer (₹)	Corporate (₹)	Small Business (₹)
Central	2,497.5	5,272.5	2,775.0
East	1,665.0	1,498.5	3,163.5
North	1,054.5	555.0	499.5
South	1,110.0	2,109.0	999.0
West	4,218.0	1,998.0	2,220.0

- Corporate dominates Central (₹5,272.5), the region always tied to the largest 5-unit order quantity, pulling Corporate's overall segment total ahead of Consumer.
- Small Business leads in East (₹3,163.5), while Consumer leads in West (₹4,218.0) — showing the segment mix is not identical across regions even though quantity is fixed by region.

- North is the smallest contributor across all three segments, consistent with its fixed 1-unit order quantity.

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14. Methodology & Data Preparation Notes

This section documents how the underlying figures in this report were derived, for reproducibility.

Data extraction

The supplied order-level PDF export was parsed into a structured table of 200 rows using pattern matching against the known field vocabulary (fixed lists of segments, regions, categories, products and payment methods), then validated by recomputing Gross Sales as Quantity × Unit Price for every row and confirming an exact match.

Derived fields

- Month — calendar month extracted from Order Date, used for the monthly trend and cumulative-sales views.
- Day of Week — weekday name extracted from Order Date, used for the order-timing view.
- Margin % — Profit ÷ Net Sales × 100, computed at the category, segment, product and overall levels.
- Cost ratio — Cost ÷ (Quantity × Unit Price) × 100, computed per product to show markup structure.

Known data-quality caveat

One row in the raw export (order ORD-30115, Desk Lamp, quantity 5) contained a corrupted Gross Sales field inconsistent with the row's other values. It was corrected to ₹200 (5 × ₹40 unit price, matching every other quantity-5 Desk Lamp order in the cycle) before aggregation, consistent with how the rest of that product's pricing behaves across the dataset.

Summary of Key Metrics

Metric	Value
Total orders	200
Date range	1 Jan 2024 – 18 Jul 2024
Total units sold	600
Total gross sales	₹33,300
Total discount given	₹1,665 (≈5.0% of gross)
Total net sales	₹31,635
Total cost	₹21,180
Total profit	₹10,455
Overall profit margin	≈33.0%
Average order value (net)	≈₹158
Top category (net sales)	Furniture (₹9,120)
Top product (net sales)	Monitor Arm (₹6,840)

Metric	Value
Highest-margin category	Office Supplies (≈47.4%)
Best month (net sales)	May 2024 (₹6,918)
Distinct customer names / unique IDs	20 / 200

Next Steps to Move from Descriptive → Predictive

- Demand forecasting by category and product once a longer, less structurally-repeating order history is available.
- Discount-elasticity analysis linking discount depth to order volume, once more than two discount tiers exist in the data.
- Customer lifetime-value and repeat-purchase analysis once orders can be reliably linked to the same customer over time.
- Regional and payment-method analysis decoupled from order quantity, once real-world regional data is layered in.
- Cohort and seasonality analysis once a full 12-month (or multi-year) order history becomes available.

This report answers what the supplied order data shows today. The next analytical layer would combine this order-level data with a longer, more organically-varied transaction history to explain why sales and profitability patterns occur, and to forecast what is likely to happen next.

Glossary

- **Gross Sales —** Quantity × Unit Price, before any discount is applied.
- **Net Sales —** Gross sales minus discount amount; the actual revenue recognized.
- **Discount % —** The percentage reduction applied to gross sales to arrive at net sales.
- **Profit —** Net sales minus cost of goods sold.
- **Margin % —** Profit expressed as a percentage of net sales.
- **Segment —** The customer type placing the order (Consumer, Corporate, or Small Business).
- **Region —** The geographic region associated with the order (North, South, East, West, Central).

Thank You

Prepared by Piyush Kumar Yadav